

Inverse Functions & Trigonometric Equations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ **1. I can **read** inverse-trigonometric notation and distinguish it from reciprocal notation.**

RELATED SKILLS

I Read inverse-trigonometric notation and distinguish it from reciprocal notation.

A Relate sine/cosecant, cosine/secant, and tangent/cotangent.

☐ **2. I can **evaluate** inverse sine, inverse cosine, and inverse tangent expressions.**

RELATED SKILLS

I Evaluate an inverse trigonometric expression exactly or approximately.

I Recall the principal-value ranges of inverse trigonometric functions.

A Express common trigonometric values exactly.

A Read inverse-trigonometric notation and distinguish it from reciprocal notation.

☐ **3. I can **explain** how restricted domains make inverse trigonometric functions possible.**

RELATED SKILLS

D Recall the principal-value ranges of inverse trigonometric functions.

A Apply the horizontal line test for invertibility.

A Exchange domain and range between a function and its inverse.

A Test whether a function is one-to-one on a domain.

☐ **4. I can **solve** trigonometric equations exactly or approximately on a specified interval.**

RELATED SKILLS

D Isolate a trigonometric expression before solving.

D Use reference angles and periodicity to find every solution in an interval.

A Check a proposed solution in the original equation.

A Determine a reference angle and its quadrant.

A Evaluate an inverse trigonometric expression exactly or approximately.

☐ **5. I can **express** all solutions of a trigonometric equation in general form.**

RELATED SKILLS

I Express a periodic solution family in general form.

D Generate and recognize coterminal angles.

A Use reference angles and periodicity to find every solution in an interval.



6. I can **evaluate compositions of trigonometric and inverse trigonometric functions.**

RELATED SKILLS

D Evaluate an inverse trigonometric expression exactly or approximately.

A Recall the principal-value ranges of inverse trigonometric functions.

A Substitute one function formula into another.

A Verify inverse functions using composition.
